

Wavelet-Triggered Deep Learning for Transient Power Quality Disturbance Detection and Classification

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Abstract—This paper presents a transient detection and classification system for power quality monitoring using a hybrid architecture combining signal processing and deep learning with a continuous streaming architecture. Unlike several existing methods that rely on pre-segmented windows, the proposed system operates directly on streaming signals using wavelet energy as detection trigger for transient localization. An event driven mechanism is employed where the convolutional neural network (CNN) runs only upon transient detection reducing unnecessary computational costs. The detected transients are classified into impulsive and oscillatory types with corresponding timestamps. Results demonstrate a detection rate of 99.99% and a classification accuracy of 99.42%.

Index Terms—Transient detection, Wavelet transform, Deep learning, Convolutional Neural Network, Power quality, Streaming signal processing, Event-driven inference.

I. INTRODUCTION

A. Motivation

Power quality disturbances (PQDs) such as impulsive and oscillatory transients are momentary but destructive, often leading to immediate equipment failure. Transient disturbances occur in power systems due to faults, switching operations, and lightning [1]. The increasing integration of renewable energy sources has intensified power quality concerns due to the intermittent nature of these sources [2]. Furthermore, modern microprocessor-based equipment is more sensitive to power quality variations than conventional loads [3]. Detection and classification of PQDs are key pillars of modern power quality monitoring systems [4]. Traditional signal processing methods are effective for detection but lack robustness in classification capability. Hence, for classification tasks artificial intelligence methods are commonly employed [5]. This paper seeks to integrate both approaches.

B. Objective

The objective of this work is to develop a hybrid signal processing and deep learning based system that:

- Detects the presence of power quality transients in a continuous signal and outputs precise timestamps using wavelet energy, eliminating the need for pre-segmentation
- Enables signal streaming via a circular buffer architecture for continuous monitoring

- Employs an event-driven inference mechanism, where the convolutional neural network (CNN) is called only upon transient detection thereby reducing computational overhead by avoiding continuous inference
- Accurately classifies transients into impulsive and oscillatory classes using a lightweight CNN suitable for embedded deployment.

II. LITERATURE REVIEW

The wavelet transform has been widely used in the field of power quality disturbance detection. Santoso *et al.* [6] established that decomposing signals at multiple resolutions enables effective localization and separation of disturbances from the underlying signal. Notably, the use of neural networks as tools for classification purposes was suggested, although this was not implemented in the paper. Wang and Chen. [7] highlighted the significance of deep learning in automatic feature extraction and thereby overcomes the drawbacks of manual feature extraction. The paper proposed a 1-D CNN that captures disturbance features with a classification accuracy exceeding 99%. However, the approach focused purely on classification and validation relied on pre-segmented windows making it unsuitable for streaming signals. Perez-Anaya *et al.* [8] proposed a methodology combining Continuous Wavelet Transform (CWT) and CNN for detection and classification of PQDs. The system, however, also operates on pre-segmented windows and does not output transient timestamps. Furthermore, the CWT serves solely to generate 2-D spectrogram images rather than as a detection mechanism. To the best of the author's knowledge, existing studies do not combine wavelet energy thresholding as a detection trigger with CNN transient type inference in a framework that supports continuous signal streaming and outputs timestamps.

III. METHODOLOGY

A. Overview

The proposed system architecture is shown in Fig. 1 and is designed for continuous monitoring. The pipeline consists of two stages: a digital signal processing (DSP)-based detection stage and a deep learning classification stage. The system accepts an input signal sampled at 100 kHz. To support streaming architecture, a circular buffer of 10,000 samples is

implemented storing a window of 100ms of data. Discrete wavelet transform (DWT) is run on the buffered signal. The db4 (member of Daubechies family) was chosen as the mother wavelet. Originally introduced in [9] and established in [6], as highly useful in power quality assessment. Following this, the approximation coefficients are removed, leaving only the detail coefficients which represent transient energy. Six levels of decomposition were chosen to narrow the a_6 approximation band to a range (0–781 Hz) that includes the 50 Hz fundamental for more precise filtering (discussed in Section VI). The signal is then reconstructed from these detail coefficients via Inverse Discrete Wavelet Transform (IDWT). To ensure that the system is robust to varying levels of noise, an adaptive threshold is set as 10 times the energy of the 99th percentile of the first 20ms baseline signal. Crucially, the first 20ms are reserved as setup time for the system during which transients are not expected to be present. Once the DWT energy exceeds the adaptive threshold, the system triggers the second stage for transient type classification. The trigger based mechanism ensures that the CNN is called only when a transient is detected, as opposed to operating it continuously. This significantly reduces the computational cost. The detected transient window is extracted from the circular buffer, resampled to a fixed length of 512 samples and fed into the CNN. Following this, the system refines the start times (for impulsive case) for higher precision. The final system output includes identified transients (if any), corresponding type, timestamps (start and end) along with confidence from the CNN. These are exported to a CSV file for record keeping.

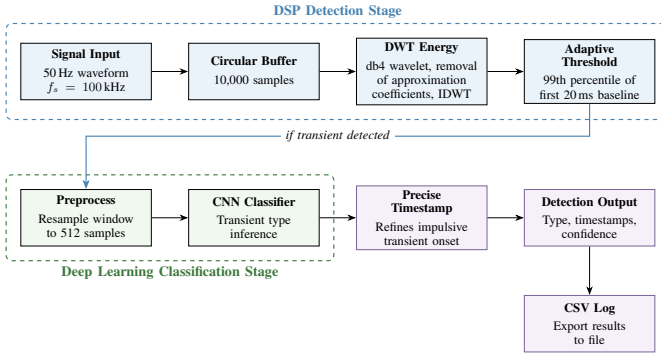


Fig. 1. Block diagram of the proposed system

B. Transient Models

The mathematical models used for generating the transient waveforms were adapted from those proposed in [10]. The parameters were configured to match impulsive transients in the microsecond to millisecond duration and low frequency oscillatory transients (less than 5 kHz) as outlined by the IEEE Std 1159-2019.

The fundamental signal is defined with an amplitude $A = 240\sqrt{2}$ V, angular frequency $\omega = 2\pi f$, where $f = 50$ Hz. The phase difference ϕ is uniformly distributed in the interval $[0, 2\pi)$. Gaussian noise $n(t)$ with mean zero and standard deviation $0.005A$ was added to the signal.

1) *Impulsive Transient*: According to IEEE 1159, “An impulsive transient is a sudden, nonpower frequency change from the nominal condition of voltage, current, or both, that is unidirectional in polarity” [1]. It is mathematically expressed as:

$$y(t) = A \cdot [\sin(\omega t + \phi) + \alpha \cdot \Psi \cdot e^{-\frac{t-t_1}{\tau}} \cdot u(t-t_1)] + n(t) \quad (1)$$

where A , ω , ϕ , and $n(t)$ retain their previously defined definitions. The term α denotes the relative magnitude of the disturbance, $u(t-t_1)$ refers to the unit step signal indicating that the transient begins at time t_1 . Lastly, ψ denotes the polarity.

2) *Oscillatory Transient*: According to IEEE 1159, “An oscillatory transient is a sudden, nonpower frequency change in the steady-state condition of voltage, current, or both, that includes both positive and negative polarity values” [1]. It is mathematically expressed as:

$$y(t) = A \cdot [\sin(\omega t + \phi) + \alpha \cdot e^{-\frac{t-t_1}{\tau}} \cdot \sin(\omega_n(t-t_1)) \cdot u(t-t_1)] + n(t) \quad (2)$$

where the additional parameter $\omega_n = 2\pi f_n$ represents the angular frequency of the oscillatory transient. The various ranges for each of the transient parameters is summarized in Table I. The parameter ranges are selected to be consistent with the characteristics of impulsive and oscillatory transients defined in IEEE Std 1159-2019.

TABLE I
TRANSIENT MODEL PARAMETERS

Parameter	Impulsive	Oscillatory
Relative magnitude (α)	[0.5, 5]	[0.1, 0.8]
Duration (d)	200 μ s–1 ms	0.3–50 ms
Frequency (f_n)	–	1–5 kHz
Polarity (Ψ)	{-1, 1}	–
Time constant (τ)	$d/4.605$	

IV. DEEP LEARNING NETWORK ARCHITECTURE

The deep learning topology is shown in Fig. 2. The input to the first convolutional layer (Conv 1) is a resampled 1-D signal window of 512 samples, which represents the detected transient isolated after the DSP stage. The network consists of three 1-D convolutional layers, of which the first two are followed by max-pooling. The output is then passed through a global average pooling (GAPool) layer and two fully connected (FC) layers. The Rectified Linear Unit (ReLU) is used as the activation function in all convolutional and the first dense layer, with a Softmax activation in the final dense layer.

A. Hyperparameters

The network was trained using the hyperparameters listed in Table II. Sparse categorical cross-entropy was used as the loss function:

$$L = - \sum y_{\text{true}} \log(y_{\text{pred}}) \quad (3)$$

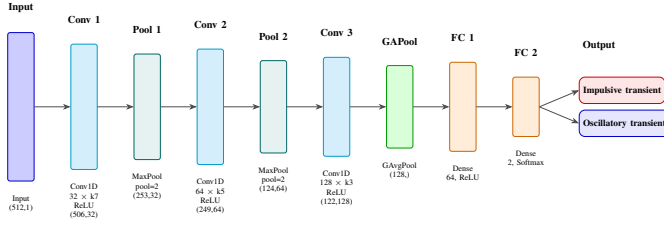


Fig. 2. Network Topology

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Number of CNN layers	3
Number of Filters	32, 64, 128
Kernel Size	7, 5, 3
Activation (Hidden Layers)	ReLU
Activation (Output Layer)	Softmax
Optimizer	Adam
Loss function	Sparse categorical cross-entropy
Epochs	20
Batch size	32
Validation split	0.2
Learning rate	0.001

B. Model Statistics

The trained model is lightweight and suitable for embedded deployment, as summarised in Table III. The int8 quantized model occupies 42.6 KB, which is achieved by representing each weight as an 8 bit integer rather than 32 bit float.

TABLE III
TRAINED MODEL STATISTICS

Property	Value
Trainable parameters	43,650
Model size (float32)	170.5 KB
Model size (int8, quantized)	42.6 KB

V. DATASET AND PERFORMANCE METRICS

A. Synthetic Training Dataset

A synthetic dataset was generated using the transient models described in Section III, comprising 20,000 samples equally distributed between the two classes. The dataset properties are summarised in Table IV. Gaussian noise was added at 0.5% of the peak amplitude A , corresponding to a Signal to Noise Ratio (SNR) of:

$$\text{SNR} = 20 \log_{10} \frac{A_{\text{signal}}}{A_{\text{noise}}} \approx 43 \text{ dB} \quad (4)$$

B. Performance Metrics

1) *Detection Metrics*: The detection stage is evaluated using three metrics. The detection rate measures the proportion of true transient events correctly identified within a ± 5 ms match window:

$$\text{Detection Rate} = \frac{TP}{TP + FN} \quad (5)$$

TABLE IV
SYNTHETIC TRAINING DATASET PROPERTIES

Property	Value
Dataset Format	NumPy (.npy)
Number of Channels	One
Total samples	20,000 (10,000 per class)
Sampling rate	100 kHz
Test signal duration	500 ms
Input length (CNN)	512 points
Noise level (SNR)	43 dB
Impulsive duration	200 μ s–1 ms
Oscillatory duration	0.3–50 ms
Oscillatory frequency	1–5 kHz

where TP denotes correctly detected transients and FN denotes missed transients. The mean absolute timing error measures the accuracy of transient onset times, reported separately for each transient type:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |t_{\text{detected},i} - t_{\text{true},i}| \quad (6)$$

The false alarm rate measures incorrect detections, where FP denotes detections reported by the system when no actual transient occurred in the signal.

$$\text{False Alarm Rate} = \frac{FP}{FP + TP} \quad (7)$$

2) *Classification Metrics*: The CNN model is evaluated using the following metrics derived from the confusion matrix shown in Fig. 3:

		True Class	
		Impulsive	Oscillatory
Predicted Class	Impulsive	TP	FP
	Oscillatory	FN	TN

Fig. 3. Confusion matrix

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

VI. ANALYSIS OF WAVELET-BASED TRIGGERING

The waveforms of the modeled transients generated using (1) and (2) are illustrated in Fig. 4 and Fig. 5, respectively.

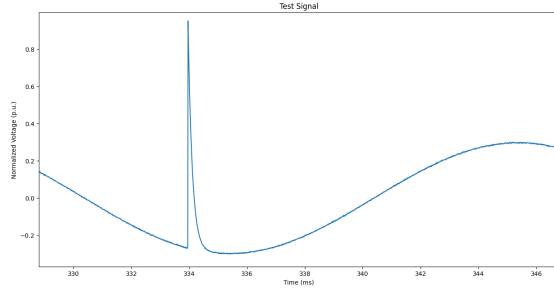


Fig. 4. Modeling of Impulsive Transient using Eq. 1.

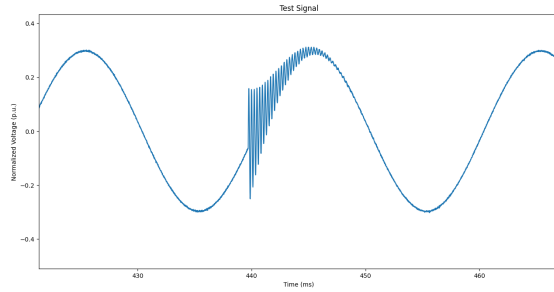


Fig. 5. Modeling of Oscillatory Transient using Eq. 2.

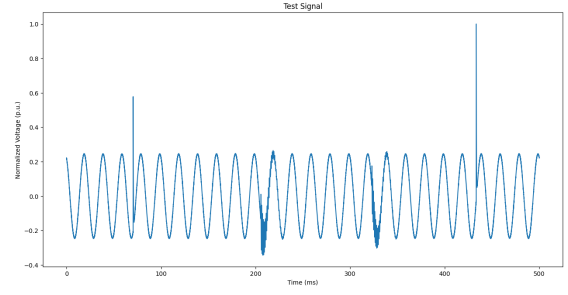


Fig. 6. Test signal as a 50Hz fundamental with various transient disturbances

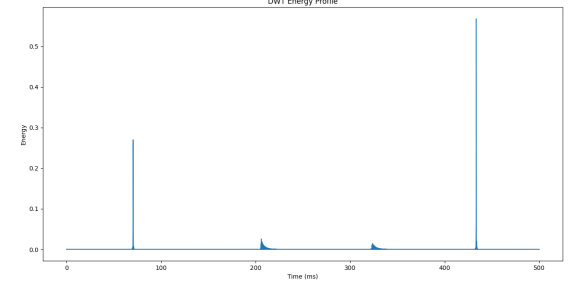


Fig. 7. DWT detail energy profile

A. Wavelet-Energy Analysis

A test signal of duration 500 ms containing two impulsive and two oscillatory transients (with added Gaussian noise) was generated, as shown in Fig. 6. The DWT energy profile (with approximation coefficients removed) in Fig. 7 confirms that the wavelet transform effectively isolates transients, producing sharp energy peaks at corresponding onset times. The results confirm that an adaptive threshold is necessary for robust detection as a fixed threshold would either miss transients entirely or trigger numerous false alarms on noise.

B. Filter Parameters: Multi-Resolution Analysis

The signal is decomposed via DWT up to Level 6. A sampling frequency of 100 kHz corresponds to a Nyquist frequency of 50 kHz. Each level splits the spectrum into approximation (a_n) and detail (d_n) coefficients and the frequency range halves at each stage. The frequency bands for each level are summarized in Table V.

TABLE V
DWT FREQUENCY BAND DECOMPOSITION ($f_s = 100$ kHz)

Level (n)	Approximation Band (a_n)	Detail Band (d_n)
1	0 – 25 kHz	25 – 50 kHz
2	0 – 12.5 kHz	12.5 – 25 kHz
3	0 – 6.25 kHz	6.25 – 12.5 kHz
4	0 – 3.125 kHz	3.125 – 6.25 kHz
5	0 – 1.5625 kHz	1.5625 – 3.125 kHz
6	0 – 781.25 Hz	781.25 Hz – 1.5625 kHz

The fundamental frequency (50 Hz) lies in the approximation band a_6 . Hence, by setting the approximation coefficients to zero and reconstructing the signal using the detail coefficients (d_1 through d_6), the transients are effectively isolated from the baseline signal.

VII. RESULTS AND DISCUSSIONS

The system was evaluated over 10,000 test signals each containing 4 injected transients (2 of each type), resulting in 40,000 total disturbances. The overall performance is summarised in Table VI.

TABLE VI
SYSTEM PERFORMANCE SUMMARY

Metric	Value
Detection rate	99.99% (39,997 / 40,000)
Classification accuracy	99.42% (39,764 / 39,997)
Mean impulsive timing error	0.015 ms
Mean oscillatory timing error	0.193 ms
False alarm rate	0.25% (99 / 39,997)

The confusion matrix for the CNN is shown in Table VII. Classification metrics derived from the confusion matrix are summarized in Table VIII.

TABLE VII
CONFUSION MATRIX (40,000 DISTURBANCES)

	Predicted Impulsive	Predicted Oscillatory
Actual Impulsive	19,767	233
Actual Oscillatory	0	19,997

TABLE VIII
CLASSIFICATION PERFORMANCE METRICS

Metric	Impulsive	Oscillatory
Precision	100%	98.85%
Recall	98.85%	100%
F1 Score	0.994	0.994
Accuracy	99.42%	

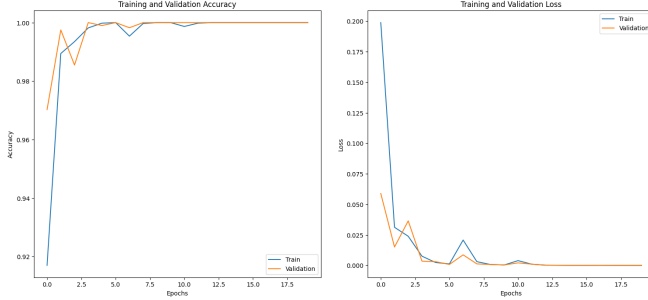


Fig. 8. Training and validation: Accuracy and Loss curves

A. Observations

The system achieved near perfect detection with only 3 missed out of 40,000 total disturbances. These missed disturbances (all oscillatory in nature) were observed to be concentrated at low relative magnitude ($\alpha \approx 0.1$) combined with short duration ($d \approx 0.3$ ms). These correspond to the edge cases of the IEEE 1159 parameter ranges. All 233 misclassifications were impulsive transients predicted as oscillatory, with no oscillatory transients misclassified. The mean timing error for impulsive transients was 0.015 ms, approximately $10\times$ lower than for oscillatory transients at 0.193 ms. Additionally, the training and validation performance of the 1D-CNN model over 20 epochs is illustrated in Fig. 8. The model converges rapidly, exceeding 99% accuracy within the first 5 epochs. The loss function was observed to drop sharply and decay to zero. Both the training and validation curves were closely aligned with one another confirming that the model was not overfitting.

VIII. CONCLUSION

This paper presented a streaming capable transient detection and classification system combining wavelet based energy triggering and CNN inference. Unlike conventional approaches that rely on pre-segmentation, the proposed method operates directly on continuous signals and provides precise transient timestamps. Results indicate that the 1D-CNN model achieves high classification accuracy while remaining lightweight, making it suitable for microcontroller deployment. Additionally, since the CNN is called only when a transient is detected, computational cost is further reduced. Future work may explore testing on real world disturbances as well as hardware monitoring.

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